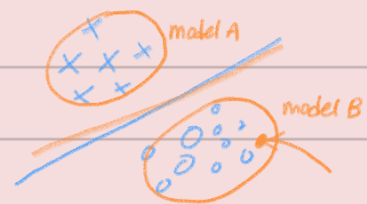


Lecture 5. Generative Learning Algorithms (GDA, Naive Bayes)

- Gaussian Discriminative Analysis (GDA)
- Generative <> Discriminative Comparison
- Naive Bayes

Discriminative $\rightarrow$ 区別에 중점 (separation)

Logistic Regression: find the decision boundary with MLE



Discriminative:

Learn $P(y|x)$ or learn $h_\theta(x) = \{0, 1\}$

Generative:

Learn $P(x|y)$ $p(y)$: class prior
 ↓ ↓
 feature class

Bayes rule:

$$P(y=1|x) = \frac{P(x|y=1)P(y=1)}{P(x)}$$

↑
 $P(x) = P(x|y=1) \cdot P(y=1) + P(x|y=0) \cdot P(y=0)$

Gaussian Discriminative Analysis (GDA)

Suppose $x \in \mathbb{R}^n$ (Prop $x_0=1$ convention)

Assume $P(x|y)$ is Gaussian
= 1D or 2D plot 안에서는 2점의 Gaussian을 그린다

$$P(z) = \frac{1}{(2\pi)^{\frac{n}{2}} |\Sigma|^{\frac{1}{2}}} \cdot \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

$$z \sim N(\vec{\mu}, \Sigma) \quad z \in \mathbb{R}^n$$

Also $E[z] = \mu$

$$E[z] = \mu$$

$$\text{COV}(z) = E[(z-\mu)(z-\mu)^T]$$

$$= Ezz^T - (Ez)(Ez)^T$$

$$P(x|y=0) = \frac{1}{(2\pi)^{\frac{n}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x-\mu_0)^T \Sigma^{-1}(x-\mu_0)\right)$$

Same covariance matrix

$$P(x|y=1) = \frac{1}{(2\pi)^{\frac{n}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x-\mu_1)^T \Sigma^{-1}(x-\mu_1)\right)$$

params: $\mu_0, \mu_1, \Sigma, \phi$
 $\mathbb{R}^n \quad \mathbb{R}^n \quad \mathbb{R}^{n \times n} \quad \mathbb{R}^{(n+1)}$

$$P(y) = \phi^y (1-\phi)^{1-y} \quad (P(y=1) = \phi)$$

Training set $\{(x^{(i)}, y^{(i)})\}_{i=1}^m$

$$\text{Joint Likelihood } \mathcal{L}(\phi, \mu_0, \mu_1, \Sigma) = \prod_{i=1}^m P(x^{(i)}, y^{(i)}; \phi, \mu_0, \mu_1, \Sigma)$$

$$= \prod_{i=1}^m P(x^{(i)}|y^{(i)}) \cdot p(y^{(i)}) \quad \leftarrow \text{Generative learning Algo}$$

Discriminative:

Conditional likelihood

$$\mathcal{L}(\theta) = \prod_{i=1}^M P(y^{(i)} | x^{(i)}; \theta)$$

indicator

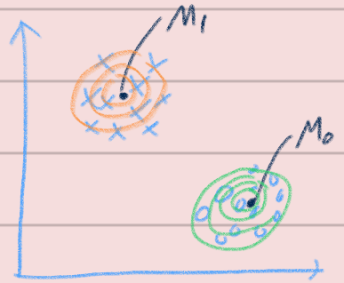
$$\begin{aligned} 1\{\text{true}\} &= 1 \\ 1\{\text{false}\} &= 0 \end{aligned}$$

MLE: Find $\phi, \mu_0, \mu_1, \Sigma$ s.t. maximize $\mathcal{L}(\phi, \mu_0, \mu_1, \Sigma) = \log \mathcal{L}(\phi, \mu_0, \mu_1, \Sigma)$

$$\phi = \frac{\sum_{i=1}^M y^{(i)}}{M} = \frac{\sum_{i=1}^M 1\{y^{(i)}=1\}}{M}$$

$$\mu_0 = \frac{\sum_{i=1}^M 1\{y^{(i)}=0\} x^{(i)}}{\sum_{i=1}^M 1\{y^{(i)}=0\}} = \text{sum of the feature vectors where } (y^{(i)}=0) \\ \text{e.g. benign tumor}$$

$$\mu_1 = \frac{\sum_{i=1}^M 1\{y^{(i)}=1\} x^{(i)}}{\sum_{i=1}^M 1\{y^{(i)}=1\}}$$



$$\Sigma = \frac{1}{M} \sum_{i=1}^M (x^{(i)} - \mu y^{(i)})(x^{(i)} - \mu y^{(i)})^T$$

$$\min (z-5)^2 = 0 \rightarrow \operatorname{argmin} \frac{(z-5)^2}{x} = 5$$

x를 최소로 만들기 위한 값

Prediction:

$$\operatorname{argmax}_y P(y|x) = \operatorname{argmax}_y \frac{P(x|y)P(y)}{\cancel{P(x)}} = \operatorname{argmax}_y P(x|y)P(y)$$

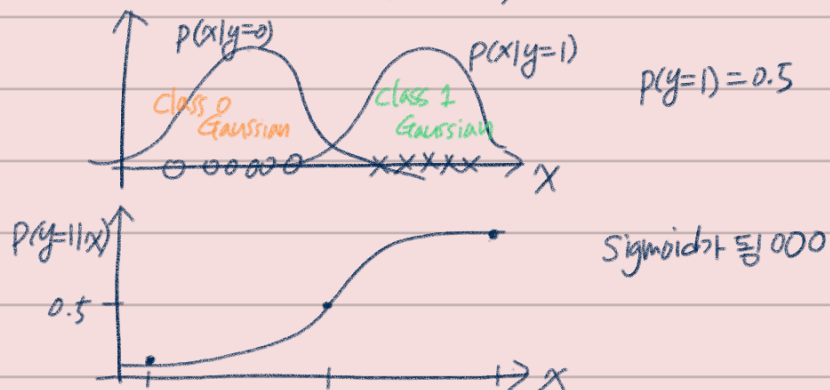
y가 가장

GDA <>

Comparison to logistic regression

For fixed $\phi, \mu_0, \mu_1, \Sigma$, let's plot $P(y=1|x; \phi, \mu_0, \mu_1, \Sigma)$ as a function of x .

$$= \frac{P(x|y=1; \dots) P(y=1; \phi)}{P(x; \phi, \mu_0, \mu_1, \Sigma)}$$



Generative approach

GDA assumes

$$x|y=0 \sim N(\mu_0, \Sigma)$$

$$x|y=1 \sim N(\mu_1, \Sigma)$$

$$y \sim \text{Ber}(\phi)$$

exponential family & natural parameters
선형 가능

implies
but the opposite is not true

~~*~~

stronger assumption

$$x|y=1 \sim \text{Poisson}(\lambda_1)$$

$$x|y=0 \sim \text{Poisson}(\lambda_0)$$

$$y \sim \text{Ber}(\phi)$$

테스트가 작을 때

$\Rightarrow P(y=1|x)$ is logistic

computationally 유리 (바람)

CS228 practical reason to use it: computationally efficient

computing mean & variances of covariance matrices is very efficient

"테스트가 작을 때도 작동하는 ML 알고리즘을 만들려는 것이 중요하다"

Discriminative approach

$$P(y=1|x) = \frac{1}{1+e^{-\theta^T x}} \quad (x_0=1)$$

(logistic)

weaker assumption

테스트가 많을 때.

큰 개수로 인해서 편향 X.

trend

performance가 더 좋음

Naive Bayes

Feature vector x ?

$$x \in \{0,1\}^n$$

$$x = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

radish
buy
CS229
z

$$x_i = 1 \text{ if word } i \text{ appears in email}$$

want to model $p(x|y)$ & $p(y)$

2^n possible values of x

$2^n - 1$ parameters ($\because x_0=1$)

Naive Bayes Assumption (conditional independence assumption)

Assume x_i s are conditionally independent to given y :

$$P(x_1, \dots, x_{1000}|y) = p(x_1|y) \cdot p(x_2|x_1, y) \cdot p(x_3|x_1, x_2, y) \dots \cdot p(x_{1000}|x_1, \dots, y)$$

$$= p(x_1|y) \cdot p(x_2|y) \dots \cdot p(x_{1000}|y)$$

Parameters:

$\phi_{j|y=1} = P(x_j=1|y=1)$ when the email is spam ($y=1$), chance of word j appears in the email

$\phi_{j|y=0} = P(x_j=1|y=0)$

$\phi_y = P(y=1)$ cost prior: $\frac{1}{m}$ the prior probability the next e-mail is a spam

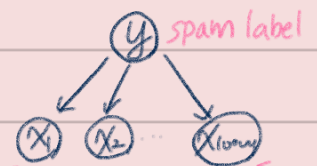
Joint Likelihood: $\mathcal{L}(\phi_y, \phi_{j|y}) = \prod_{i=1}^m P(x^{(i)}, y^{(i)}; \phi_y, \phi_{j|y})$

$$\text{MLE: } \phi_y = \frac{\sum_{i=1}^m 1\{y^{(i)}=1\}}{m}$$

$$\phi_{j|y=1} = \frac{\sum_{i=1}^m 1\{x_j^{(i)}=1, y^{(i)}=1\}}{\sum_{i=1}^m 1\{y^{(i)}=1\}}$$

the number of spam emails having word j
All the spam emails (the number of spam emails)

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whether 'buy' appears in the email
whether 'mortgage' appears in the email